

Shared Attribute Supervision for Respiratory Sound Learning Across Native Tasks

Abstract—Respiratory-sound datasets annotate related acoustic phenomena using different units, category granularities, and annotation coverage. We investigate whether compatible supervision from these datasets can support shared sound attributes while retaining their native classification targets. A shared BEATs encoder learns abnormality, Crackle, and Wheeze predictions from annotation-supported targets. A two-level decision scheme uses these predictions for ICBHI cycle-level four-class classification and SPRSound event-level Normal/Adventitious classification. The study distinguishes the reuse of attribute information across annotation conditions from the effect of the final task readout. Preliminary results from three ICBHI-test-selected runs achieve an ICBHI Score of $61.17 \pm 0.31\%$ and a SPRSound official Score of $90.70 \pm 0.34\%$ with one joint model per run. These results establish a native-task operating point. Whether cross-source supervision compensates for missing fine-grained labels, and how much the two-level decision contributes, remain to be determined by the specified matched ablations.

Index Terms—respiratory sounds, shared supervision, heterogeneous annotations, acoustic attributes, hierarchical decoding

I. INTRODUCTION

Respiratory-sound datasets provide complementary observations of related acoustic events, but their annotations answer different questions. ICBHI assigns Normal, Crackle, Wheeze, or Both to complete respiratory cycles. SPRSound annotates individual sound events and defines both coarse Normal/Adventitious classification and finer event categories [1], [2]. Differences in annotation units, category granularity, and coverage therefore accompany the acoustic variation across data sources. Using these datasets together requires deciding which supervision can be shared and which distinctions each task must retain.

A shared acoustic representation offers a way to connect these tasks. Crackle and Wheeze annotations can supervise the same attributes across sources, while an abnormality label supplies a coarser distinction. However, these training signals carry different amounts of information. An abnormality label alone does not specify a fine sound attribute, and merging fine categories can remove distinctions required by another task. The central question is whether compatible attribute supervision can preserve useful acoustic information across these annotation conditions and support the original classification targets through a common model.

Several lines of work provide the foundations for this question. SPRSound includes an ICBHI/SPRSound data-fusion benchmark [2], and LungMix unifies respiratory labels into a common four-class space for single-source generalization [3]. OPERA learns shared respiratory representations and

evaluates them with separate downstream predictors [4]. In heterogeneous supervised learning, universal taxonomies map shared predictions back to dataset-specific labels [5], and DCASE 2024 combines audio datasets with label mapping and missing-target masking [6]. Respiratory attribute classification also has close precedents: PC-MCL uses explicit multi-label targets and deterministic four-class conversion [7], while OCAD develops event factors and structured compositional decoding [8]. Our focus is the specific relationship between shared attribute supervision, annotation conditions, and native respiratory-task readouts.

We study joint ICBHI cycle-four-class and SPRSound event-binary learning. The model shares abnormality, Crackle, and Wheeze predictions, supervises each only where the source annotation supports a target, and applies a two-level decision scheme. SPRSound reads abnormality directly; ICBHI combines it with the two attributes. This separates the supervision available during learning from the output requested by a task. In particular, a binary endpoint can still use fine labels available in its training data.

The draft is organized around three contributions:

- 1) An annotation-aware joint learning framework that shares respiratory attributes across cycle and event annotations and returns their native classification targets through a two-level decision scheme.
- 2) A characterization of cross-source attribute reuse under coarse and partially available annotations, examining whether another source’s supervision supports fine-grained recognition.
- 3) A supervision-matched analysis of the separate effects of attribute information and task decoding on native-task performance and class-specific trade-offs.

Draft note. Contributions 2 and 3 are intended empirical contributions. Their findings await the annotation and classification ablations in Sections IV-C and IV-D. The present numerical results support the joint prediction capability.

II. DATA AND SHARED SUPERVISION

A. Native Prediction Units and Targets

The core study uses two presegmented classification tasks. ICBHI provides respiratory cycles labeled Normal, Crackle, Wheeze, or Both [1]; Both denotes the presence of the two attributes within the cycle. SPRSound provides event annotations, with Task1-1 asking whether an event is Normal or Adventitious [2]. Each native unit receives one prediction. This task definition preserves cycle and event labels without introducing a new temporal-localization objective.

Native annotation	a	c	w
Normal (ICBHI / SPRSound)	0	0	0
Crackle / Coarse or Fine Crackle	1	1	0
Wheeze (ICBHI / SPRSound)	1	0	1
Both / Wheeze+Crackle	1	1	1
Rhonchi or Stridor (SPRSound)	1	–	–

Fig. 1. Shared supervision from native labels. A dash marks an unavailable target. Fine SPRSound event labels contribute attributes even though its primary evaluation target is binary.

We use the ICBHI official recording split: 4,142 training and 2,756 test cycles. The test supports are 1,579 Normal, 649 Crackle, 385 Wheeze, and 143 Both. This official split is not strictly patient-held-out. SPRSound evaluation uses the BioCAS2022 official inter-subject partition: 1,429 events, comprising 1,040 Normal and 389 Adventitious examples. Its intra-subject test and seven-class Task1-2 are separate endpoints.

B. Shared Attributes and Annotation Coverage

Let a , c , and w denote abnormality, Crackle, and Wheeze. ICBHI labels determine all three targets. Compatible SPRSound event labels supply the same targets, with Coarse and Fine Crackle merged at the attribute level. Rhonchi and Stridor supervise abnormality; their C/W targets are left unavailable in our mapping. An unavailable target supplies neither positive nor negative classification supervision. Figure 1 shows the shared target space.

The retained attributes must be sufficient to express the selected task. They express ICBHI’s four categories and SPRSound’s binary target, but do not retain the Coarse/Fine Crackle distinction required for SPRSound seven-class classification. Native-task preservation here refers to the specified units and category definitions; performance retention is assessed against single-task references.

C. Input Construction

Recordings are converted to mono and resampled to 16 kHz. The inherited pipeline applies a 1/16-s linear fade at recording boundaries before extracting each annotated unit. Units longer than 5 s retain their first 5 s; shorter units are repeated to 5 s and receive an end fade. The two datasets therefore share an input format while retaining their original supervision units. This preprocessing does not establish that every long unit’s acoustic events are retained.

III. METHOD

A. Shared Acoustic Representation and Attributes

Figure 2 separates the acoustic input path from the annotation path. We initialize the encoder with BEATs iter3+ AS2M [9] and fine-tune it jointly on the two datasets. Its 768-dimensional token features are mean-pooled and linearly projected to $h_i \in \mathbb{R}^{256}$. Three parallel linear predictors produce

$$\begin{aligned} p_{ia} &= \text{softmax}(W_a h_i + b_a)_{\text{Abnormal}}, \\ p_{ic} &= \sigma(W_c h_i + b_c), \quad p_{iw} = \sigma(W_w h_i + b_w). \end{aligned} \quad (1)$$

The abnormality predictor has two logits; each attribute predictor has one. All three update the shared representation. The two-level structure is used for task decisions: the C/W probabilities are marginal predictions, with no explicit parent-child probability-consistency constraint.

B. Learning from Available Supervision

For sample i and node $k \in \{a, c, w\}$, $m_{ik} = 1$ when its source annotation supports target y_{ik} . For batch B , let $B_k = \{i \in B : m_{ik} = 1\}$ and $K_B = \{k : |B_k| > 0\}$. The implemented classification objective averages over available samples within each node:

$$\mathcal{L}_{\text{cls}}(B) = \frac{1}{|K_B|} \sum_{k \in K_B} \frac{1}{|B_k|} \sum_{i \in B_k} \ell_k(p_{ik}, y_{ik}), \quad (2)$$

where ℓ_a is two-class cross-entropy and ℓ_c, ℓ_w are binary cross-entropy. Dataset-homogeneous batches contribute equally many updates per epoch. Thus supported labels from either source train the shared attributes without requiring every sample to supervise every node.

We retain PAFA’s patient cohesion-separation and global patient alignment objectives [10]:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 50\mathcal{L}_{\text{PCSL}} + 0.0005\mathcal{L}_{\text{GPAL}}. \quad (3)$$

An auxiliary attention-based projection supplies their features, with patient identifiers qualified by dataset. This auxiliary projection is used during training and does not enter the native classification readouts.

C. Two-Level Native Task Decisions

The Normal/Abnormal argmax defines $\hat{a}$; its ties select Normal. Attribute decisions are $\hat{c} = \mathbb{I}[p_c \geq \tau_c]$ and $\hat{w} = \mathbb{I}[p_w \geq \tau_w]$. SPRSound directly uses $\hat{a}$ for Normal/Adventitious classification. ICBHI assigns Normal if $\hat{a} = 0$. For an abnormal decision, two positive attributes yield Both and one positive attribute yields its corresponding class. If neither is positive, the larger $p_k - \tau_k$ margin selects Crackle or Wheeze, with ties assigned to Crackle.

For the reported runs, the shared C/W thresholds independently maximize the equal-dataset mean of eligible-node binary F1 on internal validation data. Candidates are the observed probabilities together with 0 and 1; exact ties select the larger threshold. This readout returns task labels from the same attribute scores. Its contribution can therefore be studied separately from representation learning by fixing the scores and changing only the decision rule.

IV. EVALUATION

The evaluation addresses three questions: whether a shared model supports both native tasks; whether attribute information transfers when annotation coverage changes; and how much the final classification design contributes. The current results answer the first at the level of a joint operating point. The latter two require the specified ablations.

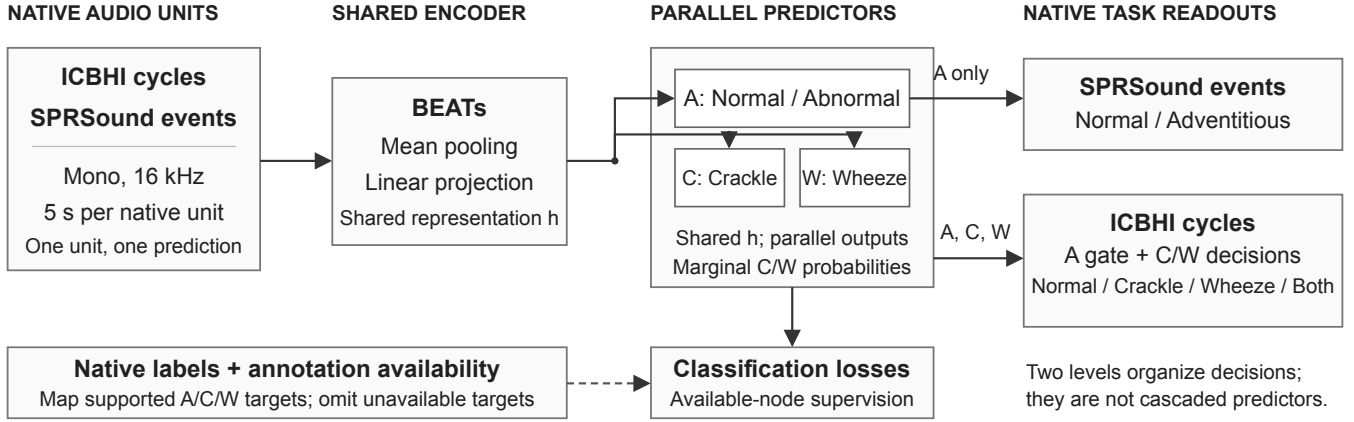


Fig. 2. Shared attributes with native task readouts. Audio from each native unit enters the same encoder and three parallel predictors. Labels and availability masks enter the training losses. SPRSound uses the abnormality decision; ICBHI combines abnormality and C/W decisions. The semantic decision hierarchy does not imply that the C/W predictors take the abnormality output as their input.

A. Protocol and Comparison Sources

The joint model uses full BEATs fine-tuning, batch size 32, Adam with learning rate 5×10^{-5} and weight decay 10^{-6} , a 50-epoch cosine schedule with minimum learning-rate ratio 0.001, and parameter EMA 0.5. Each epoch has 326 updates, divided equally between the sources. SpecAugment, mean-variance normalization, and HF auxiliary supervision are disabled.

Internal ICBHI validation uses the first fold of a shuffled five-fold stratified patient-group split, controlled by the run seed. SPRSound’s 6,656 official training events are divided into 5,219 fitting and 1,437 validation events, with 194 and 49 patient groups. These data set attribute thresholds. In the existing runs, checkpoint selection and patience-10 early stopping instead use ICBHI official-test Score. The formal seeds 0, 1, and fresh 42 select epochs 15, 11, and 17; SPRSound official inter is evaluated once per selected checkpoint. The resulting mean and sample standard deviation describe a test-selected benchmark. New matched comparisons will use a common validation-based selection protocol, with its split, criterion, and training budget fixed before execution.

For ICBHI, Sp is Normal recall and Se is the fraction of abnormal cycles assigned to their correct abnormal subclass. Its Score is $(Sp + Se)/2$; confusing Crackle with Wheeze therefore reduces Se . For SPRSound Task1-1, Se and Sp are Adventitious and Normal recall. Its official Score is $(AS + HS)/2$, where $AS = (Se + Sp)/2$ and $HS = 2SeSp/(Se + Sp)$.

The frozen-encoder references in Table I come from the single-dataset baseline report. They use 16-kHz audio, 2-s windows with 1-s hop and tail coverage, zero-padding, a 256-dimensional projector and native head, batch size 8, 50 epochs, and seed 42. ICBHI uses official-test Score selection; SPRSound intra/inter runs are selected separately. The source checkpoint rows are local evaluations of ICBHI-specialized models, not the corresponding published multi-run means.

TABLE I
EXISTING NATIVE-TASK RESULTS, IN PERCENT. SINGLE-DATASET ROWS USE SEPARATELY TRAINED MODELS FOR THE TWO COLUMNS; THE JOINT ROW USES ONE MODEL PER SEED. FROZEN-ENCODER VALUES ARE REPORTED SINGLE-RUN REFERENCES. RECIPES AND SELECTION PROCEDURES DIFFER, SO THE ROWS ARE NOT MATCHED ABLATIONS.

Model / training setting	ICBHI Score	SPRSound inter Official Score
<i>Single-dataset frozen encoders: seed 42</i>		
AST	51.22	89.99
BEATs	51.37	92.00
PANNs	51.97	91.31
OPERA-CT	38.61	79.45
HeAR	48.98	88.12
<i>ICBHI source checkpoint references</i>		
PAFA [10]	64.14	–
SG-SCL [11]	60.98	–
Patch-Mix [12]	62.17	–
<i>Joint full fine-tuning: three runs</i>		
Shared attributes	61.17 ± 0.31	90.70 ± 0.34

These references provide context for the native tasks, with training and input differences made explicit.

B. Native-Task Performance with a Shared Model

The joint model obtains ICBHI Score $61.17 \pm 0.31\%$ and SPRSound official Score $90.70 \pm 0.34\%$. Its ICBHI Sp and Se are $74.98 \pm 5.69\%$ and $47.35 \pm 5.35\%$. Class recall is $60.20 \pm 8.48\%$ for Crackle, $29.61 \pm 2.08\%$ for Wheeze, and $36.83 \pm 11.37\%$ for Both. SPRSound Normal and Adventitious recall are $84.29 \pm 1.36\%$ and $97.60 \pm 1.21\%$, respectively. Thus the shared model supplies both requested outputs, with uneven recognition of the fine ICBHI categories. The small ICBHI Score variation does not imply equally stable class behavior.

The frozen BEATs references obtain 51.37% on ICBHI and 92.00% on SPRSound, whereas the single-source PAFA checkpoint reaches 64.14% on ICBHI. These different reference points do not establish a sharing gain or cost: they

also differ in fine-tuning, inputs, losses, and selection. A matched joint-versus-single-source comparison is needed to assess performance retention and the value of adding another source’s supervision.

C. Attribute Reuse under Annotation Changes

To test what shared supervision contributes, the proposed annotation study keeps the joint audio and architecture fixed while withholding SPRSound C/W training targets, retaining its native abnormality supervision. Evaluation then asks whether ICBHI attribute supervision supports C/W recognition on compatible SPRSound labels, alongside its native binary task. Source-only attribute learning and the complete-label joint model provide references for transfer and available supervision, respectively. Attribute metrics use only samples whose annotations support the corresponding target.

This setting varies the information available for training, rather than the native evaluation labels. The target source still participates in coarse-task training. To isolate attribute information, the weights of retained losses must remain fixed: deleting C/W from the active-node average in (2) would otherwise also increase the relative weight of abnormality supervision. The exact validation-based selection rule and attribute evaluation protocol remain to be fixed for these runs.

Draft note. Annotation-ablation results are pending. The current draft does not establish compensation for missing fine labels or cross-source attribute reuse under reduced annotation coverage.

D. Separating Classification and Supervision Effects

Two comparisons address the two-level design. First, a shared encoder with native ICBHI-four-class and SPRSound-binary heads should have access to the same fine attribute annotations, for example through matched auxiliary attribute supervision. Comparing this model with the shared-attribute model assesses the classification interface at matched information content. A binary-only SPRSound head without attribute supervision would change both the available information and the classification design.

Second, fixing the shared model’s attribute scores allows a direct readout comparison. C/W-only four-class conversion maps two negative attributes to Normal, two positive attributes to Both, and a single positive attribute to its class. Comparing this rule with the abnormality-gated rule isolates changes in the final decisions. Both rules require validation-based threshold selection. Since the encoder and scores stay fixed, this comparison cannot be interpreted as a change in learned acoustic features.

Draft note. The comparisons in Table II specify questions and controls, not completed findings. Their results will determine whether two-level decoding improves a task, changes its operating point, or adds little beyond the shared supervision.

E. Scope of Additional Dataset Evidence

Existing HF Lung and KAUH evaluations use fixed selected checkpoints and compatible post-hoc readouts [13], [14]. The

TABLE II
ABLATIONS REQUIRED BY THE STORYLINE. RESULTS ARE PENDING; THE EXISTING JOINT BENCHMARK DOES NOT FILL THESE COMPARISON ROWS.

Comparison	Question answered
Joint vs. each single source	Value of additional source supervision
Full vs. coarse SPR supervision	Value and reuse of fine attributes
Shared attributes vs. native heads, same labels	Effect of classification interface
Two-level vs. C/W-only readout, fixed scores	Effect of the final decision rule

baseline report instead uses HF temporal multi-label prediction and KAUH recording-level nine-class classification. These are different tasks. They motivate broader study but do not demonstrate that the present three-attribute model restores those datasets’ full native targets. The main argument therefore remains centered on the two specified native tasks and the annotation/readout comparisons above.

V. CONCLUSION

We study shared respiratory-sound attribute supervision as a link between heterogeneous annotations and native classification tasks. The framework uses supported abnormality and C/W targets to train one acoustic model, then returns ICBHI cycle-four-class and SPRSound event-binary predictions through a two-level decision scheme. Existing results demonstrate this joint prediction capability and identify uneven fine-category recognition. The remaining empirical questions concern how attribute information transfers when fine labels are unavailable and what the decision hierarchy adds at matched supervision. The specified ablations will determine these effects and the resulting scope of the empirical contributions.

REFERENCES

- [1] B. M. Rocha *et al.*, “An open access database for the evaluation of respiratory sound classification algorithms,” *Physiological Measurement*, vol. 40, no. 3, p. 035001, 2019.
- [2] Q. Zhang *et al.*, “SPRSound: Open-source SJTU paediatric respiratory sound database,” *IEEE Transactions on Biomedical Circuits and Systems*, vol. 16, no. 5, pp. 867–881, 2022.
- [3] S. Ge, W. Zhang, S. Xie, B. Yan, and Z. Wang, “LungMix: A mixup-based strategy for generalization in respiratory sound classification,” in *ICASSP 2025—2025 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.
- [4] Y. Zhang *et al.*, “Towards open respiratory acoustic foundation models: Pretraining and benchmarking,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 27 024–27 055.
- [5] P. Bevandić, M. Oršić, I. Grubišić, J. Šarić, and S. Šegvić, “Multi-domain semantic segmentation with overlapping labels,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 2615–2624.
- [6] S. Cornell *et al.*, “DCASE 2024 task 4: Sound event detection with heterogeneous data and missing labels,” arXiv:2406.08056, 2024.
- [7] S. G. Jeong and S.-E. Kim, “PC-MCL: Patient-consistent multi-cycle learning with multi-label bias correction for respiratory sound classification,” in *ICASSP 2026—2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 2665–2669.
- [8] X. Zhang, W. Zhao, and H. Liang, “OCAD: Overlap composition and class-atom decoding for respiratory-sound classification,” *Applied Sciences*, vol. 16, no. 13, p. 6832, 2026.

- [9] S. Chen *et al.*, “BEATs: Audio pre-training with acoustic tokenizers,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 5178–5193.
- [10] S. G. Jeong and S. E. Kim, “Patient-aware feature alignment for robust lung sound classification: Cohesion-separation and global alignment losses,” in *Interspeech 2025*, 2025, pp. 1018–1022.
- [11] J.-W. Kim, S. Bae, W.-Y. Cho, B. Lee, and H.-Y. Jung, “Stethoscope-guided supervised contrastive learning for cross-domain adaptation on respiratory sound classification,” in *ICASSP 2024—2024 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2024, pp. 1431–1435.
- [12] S. Bae *et al.*, “Patch-mix contrastive learning with audio spectrogram transformer on respiratory sound classification,” in *Interspeech 2023*, 2023, pp. 5436–5440.
- [13] F.-S. Hsu *et al.*, “Benchmarking of eight recurrent neural network variants for breath phase and adventitious sound detection on a self-developed open-access lung sound database—HF_Lung_V1,” *PLOS ONE*, vol. 16, no. 7, p. e0254134, 2021.
- [14] M. Fraiwan, L. Fraiwan, B. Khassawneh, and A. Ibnian, “A dataset of lung sounds recorded from the chest wall using an electronic stethoscope,” *Data in Brief*, vol. 35, p. 106913, 2021.